Code-Layout, Readability and Reusability

Date	25 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform 4-space tabs used throughout Python and HTML code	Yes	PEP 8 compliant
2	Proper File Structure	Files organized into templates/, static/, model files, and app.py	Yes	Clear separation of concerns
3	Meaningful Variable Names	Variable like Nitrogen, Phosphorous, Temperature used instead of x1, x2	Yes	Self-documenting code
4	Function / Method Names	Functions named descriptively: predict(), findyourcrop(), home()	Yes	Follows Python naming conventions
5	Code Comments	Inline comments explaining ML pipeline steps, route handlers, and data processing logic	Yes	Comments present in Jupyter notebook and app.py
6	Modular Design	Separate modules for ML training (Jupyter), web app (app.py), and frontend (HTML templates)	Yes	Model training separate from deployment
7	No Redundant Code	Unused imports and duplicate code removed during code review	Yes	Clean codebase
8	Error Handling	Try-except blocks for model loading, input validation, and prediction errors	Yes	Graceful error messages displayed to user

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	ML Model (model.pkl)	Python / Scikit-learn	Flask prediction route, Jupyter testing	High
2	Data preprocessing pipeline	Python / Pandas	Training notebook, future dataset updates	High
3	HTML base template structure	HTML / Bootstrap	Home, About, FindYourCrop pages	High
4	Input validation logic	Python / Flask	Prediction route handler	Medium
5	Visualization functions	Python / Matplotlib, Seaborn	Analysis notebook, documentation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Well-organized with clear folder structure
Readability	4	Meaningful names and consistent formatting
Reusability	4	Model and preprocessing pipeline easily reusable
Documentation / Comments	4	Comprehensive comments in notebook and backend
Overall Score	4	